



MILESTONE -4

Infosys ElectViz Election Data
Visualization

**INDIAN GERNAL ELECTION – 2024
DATASET FROM KAGGLE**

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Introduction to Indian General Election 2024

The Indian General Election 2024 is one of the largest democratic exercises in the world, involving millions of voters across diverse states and constituencies. Conducted to elect members of the Lok Sabha, the election determines the central government of India. With 543 constituencies, multiple political parties, and national alliances, the election generates a vast amount of structured data related to candidates, parties, votes, and constituencies.

Analysing such large-scale election data helps in understanding voting patterns, regional preferences, alliance performance, and electoral competitiveness. Traditional analysis methods are insufficient to handle this volume and complexity of data. Hence, modern data analytics tools like Microsoft Power BI are used to transform raw election data into meaningful insights.

This project focuses on analysing the Indian General Election 2024 data using Power BI. The aim is to visualize results at national, state, and constituency levels in an interactive manner. By using dashboards, maps, charts, and DAX calculations, users can easily explore winning alliances, candidates, vote shares, and margins. This project demonstrates how data visualization can enhance transparency, improve understanding of democratic processes, and support data-driven decision-making.

2. Project Objectives & Scope

The main objective of this project is to analyse and visualize the Indian General Election 2024 results using Power BI. The project aims to convert complex election datasets into interactive dashboards that provide clear and meaningful insights. One of the primary goals is to identify the winning alliance at the national and state levels and to analyse seat distribution among political parties. Another objective is to perform constituency-level analysis to identify winning

candidates, runner-up candidates, vote shares, and victory margins. The project also focuses on understanding regional variations in voting behaviour and alliance dominance across different states.

The scope of the project includes data cleaning, data modelling, creation of calculated columns and measures using DAX, and dashboard development. It does not include predictive modelling or voter sentiment analysis. The analysis is limited to officially available election result data.

This project is useful for students, analysts, and beginners in Power BI who want hands-on experience with real-world datasets. It also helps in understanding how data analytics tools can be applied in governance, politics, and public data analysis.

3. Dataset Overview

The dataset used in this project consists of multiple tables related to the Indian General Election 2024. These tables contain structured information about candidates, political parties, constituencies, states, and vote counts. The data is organized in a tabular format suitable for analytical processing.

Key tables include PartyWiseResults, ConstituencyWiseResults, ConstituencyWiseDetails, and States. The PartyWiseResults table contains party names and their overall performance. The ConstituencyWiseResults table stores constituency-level winning information. The ConstituencyWiseDetails table includes detailed candidate-level data such as total votes and vote share percentages. The States table provides geographical mapping for state-level analysis.

Each table contains unique identifiers such as Party ID, Constituency ID, and State ID, which help in creating relationships between tables. The dataset includes both numerical data (votes, percentages) and categorical data (party names, candidate names, alliances).

Overall, the dataset is comprehensive and suitable for multi-level analysis,

enabling insights at national, state, and constituency levels using Power BI.

4. Data Sources

The data used in this project is sourced from publicly available election result datasets released by official election authorities and open data platforms. These sources provide reliable and verified information related to election outcomes, making the dataset trustworthy for analysis.

The datasets typically include detailed information such as candidate names, party affiliations, total votes received, vote share percentages, and constituency details. State and constituency boundaries are also included to support geographical analysis.

Before importing the data into Power BI, the datasets were reviewed to ensure completeness and consistency. Any missing or duplicate records were identified during the data cleaning phase. Since the data is publicly available, it is suitable for academic and learning purposes.

Using authentic data sources ensures accuracy in analysis and improves the credibility of the insights generated through the dashboards.

5. Data Cleaning

Data cleaning is a critical step in the data analysis process. Raw election data often contains inconsistencies such as missing values, inconsistent party names, extra spaces, and duplicate records. These issues can lead to incorrect analysis if not handled properly.

In this project, data cleaning was performed using Power Query in Power BI. Text functions such as TRIM and CLEAN were used to remove unnecessary spaces and special characters. Party names were standardized to ensure consistency across tables. Blank or null values were reviewed and handled appropriately.

Duplicate records were checked using unique identifiers like Constituency ID and Candidate Name. Columns with irrelevant or redundant information were removed to optimize model performance.

Proper data cleaning ensured that the dataset was accurate, consistent, and ready for reliable analysis and visualization.

6. Data Modelling

Data modelling is the process of organizing data into related tables to enable efficient analysis. In this project, a structured data model was created using Power BI's relationship view. Fact and dimension tables were identified to support analytical queries.

The ConstituencyWiseDetails table acts as the main fact table, containing measurable values such as votes and vote share. Dimension tables like States and PartyWiseResults provide descriptive information.

Relationships were created using common keys such as Constituency ID and Party ID. Most relationships follow a one-to-many structure, ensuring proper filter flow across visuals.

A well-designed data model improves performance, reduces ambiguity, and ensures accurate DAX calculations across dashboards.

7. Star Schema Design

The star schema is a widely used data modeling technique in business intelligence. In this project, a star schema design was implemented to simplify analysis and improve performance.

At the center of the schema is the fact table, ConstituencyWiseDetails, which contains quantitative data such as total votes and vote percentages. Surrounding this fact table are dimension tables like States, Parties, and Constituencies.

This design allows slicers and filters to work efficiently across visuals. It also

simplifies DAX formulas by reducing complex joins and relationships.

Using a star schema makes the model scalable and easier to understand, which is especially important for large datasets like election results.

8. DAX Fundamentals

DAX (Data Analysis Expressions) is a formula language used in Power BI for creating calculated columns and measures. It allows users to perform advanced calculations using data model context.

Calculated columns are evaluated row by row and stored in the model, while measures are calculated dynamically based on filter context. Functions like CALCULATE, FILTER, MAX, COUNTROWS, and SELECTEDVALUE are commonly used.

In this project, DAX is used to calculate winning candidates, alliances, vote shares, and seat counts. Understanding filter context and row context is essential for writing correct DAX formulas.

DAX plays a crucial role in transforming raw data into analytical insights.

1.

Party Alliance =

IF(

PartyWiseResults[Party] = "Bharatiya Janata Party - BJP" ||

PartyWiseResults[Party] = "Telugu Desam - TDP" ||

PartyWiseResults[Party] = "Janata Dal (United) - JD(U)" ||

PartyWiseResults[Party] = "Shiv Sena - SHS" ||

PartyWiseResults[Party] = "AJSU Party - AJSUP" ||

PartyWiseResults[Party] = "Apna Dal (Soneylal) - ADAL" ||

PartyWiseResults[Party] = "Asom Gana Parishad - AGP" ||

PartyWiseResults[Party] = "Hindustani Awam Morcha (Secular) - HAMS" ||

PartyWiseResults[Party] = "Janasena Party - JnP" ||

```

PartyWiseResults[Party] = "Janata Dal (Secular) - JD(S)" ||
PartyWiseResults[Party] = "Lok Janshakti Party(Ram Vilas) - LJPRV" ||
PartyWiseResults[Party] = "Nationalist Congress Party - NCP" ||
PartyWiseResults[Party]= "Rashtriya Lok Dal - RLD" ||
PartyWiseResults[Party] = "Sikkim Krantikari Morcha - SKM",
"NDA",
IF(
    PartyWiseResults[Party] = "Indian National Congress - INC" ||
    PartyWiseResults[Party] = "Aam Aadmi Party - AAP" ||
    PartyWiseResults[Party] = "All India Trinamool Congress - AITC" ||
    PartyWiseResults[Party] = "Bharat Adivasi Party - BHRTADVSI" ||
    PartyWiseResults[Party]= "Communist Party of India (Marxist) - CPI(M)"
||
    PartyWiseResults[Party] = "Communist Party of India (Marxist-Leninist)
(Liberation) - CPI(ML)(L)" ||
    PartyWiseResults[Party] = "Communist Party of India - CPI" ||
    PartyWiseResults[Party] = "Dravida Munnetra Kazhagam - DMK" ||
    PartyWiseResults[Party] = "Indian Union Muslim League - IUML" ||
    PartyWiseResults[Party] = "Jammu & Kashmir National Conference -
JKN" ||
    PartyWiseResults[Party] = "Jharkhand Mukti Morcha - JMM" ||
    PartyWiseResults[Party] = "Kerala Congress - KEC" ||
    PartyWiseResults[Party] = "Marumalarchi Dravida Munnetra Kazhagam -
MDMK" ||
    PartyWiseResults[Party] = "Nationalist Congress Party Sharadchandra
Pawar - NCPSP" ||
    PartyWiseResults[Party] = "Rashtriya Janata Dal - RJD" ||
    PartyWiseResults[Party] = "Rashtriya Loktantrik Party - RLTP" ||
    PartyWiseResults[Party] = "Revolutionary Socialist Party - RSP" ||

```

```

PartyWiseResults[Party] = "Samajwadi Party - SP" ||
PartyWiseResults[Party] = "Shiv Sena (Uddhav Balasaheb Thackrey) -
SHSUBT" ||
PartyWiseResults[Party] = "Viduthalai Chiruthaigal Katchi - VCK",
"I.N.D.I.A.",
"OTHER"
)
)

```

```

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=====
=====

```

2.

Winning Alliance =

VAR NDA_Seat_Count =

```

    CALCULATE(
        COUNT('ConstituencyWiseResults'[Winning Candidate]),
        'PartyWiseResults'[Party Alliance] = "NDA"
    )

```

VAR INDIA_Seat_Count =

```

    CALCULATE(
        COUNT('ConstituencyWiseResults'[Winning Candidate]),
        'PartyWiseResults'[Party Alliance] = "I.N.D.I.A."
    )

```


RETURN

IF(

NDA_Seat_Count >= INDIA_Seat_Count,

"NDA",

"I.N.D.I.A."

)

Selected State Name =

CALCULATE(

SELECTEDVALUE(states[State], "No State Selected"),

FILTER(

ConstituencyWiseResults,

ConstituencyWiseResults[Constituency ID] =

MAX(ConstituencyWiseDetails[Constituency ID])

)

)

=====

=====

Winning Candidate =

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

RETURN

CALCULATE(

MAX('ConstituencyWiseDetails'[Candidate]),

'ConstituencyWiseDetails'[Total Votes] = MaxVotes

)

=====

=====

=====

Winning Party Name =

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

RETURN

CALCULATE(

MAX('ConstituencyWiseDetails'[Party]),

'ConstituencyWiseDetails'[Total Votes] = MaxVotes

)

=====

=====

=====

Winning Candidate Total Votes =

"Total Votes:" & " " & MAX('ConstituencyWiseDetails'[Total Votes])

=====

=====

Winning Candidate Vote Share % =

"Vote Share:" & " " & MAX('ConstituencyWiseDetails'[% of Votes]) & " %"

=====

=====

=====

Runner Up Winning Candidate =

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

VAR SecondMaxVotes =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[Total Votes] < MaxVotes

),

'ConstituencyWiseDetails'[Total Votes]

)

RETURN

CALCULATE(

MAX('ConstituencyWiseDetails'[Candidate]),

```
'ConstituencyWiseDetails'[Total Votes] = SecondMaxVotes
)
```

```
=====
=====
=====
```

Runner UP Winning Party =

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

VAR SecondMaxVotes =

```
MAXX(
    FILTER(
        'ConstituencyWiseDetails',
        'ConstituencyWiseDetails'[Total Votes] < MaxVotes
    ),
    'ConstituencyWiseDetails'[Total Votes]
)
```

RETURN

```
CALCULATE(
    MAX('ConstituencyWiseDetails'[Party]),
    'ConstituencyWiseDetails'[Total Votes] = SecondMaxVotes
)
```

```
=====
```

=====

=====

Runner Up Winning Candidate Total Votes =

"Total Votes: " &

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

VAR SecondMaxVotes =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[Total Votes] < MaxVotes

),

'ConstituencyWiseDetails'[Total Votes]

)

RETURN

IF(

ISBLANK(SecondMaxVotes),

"No second max",

SecondMaxVotes

)

=====

=====

=====

```

Runner Up Winning Candidate Vote Share % =
"Vote Share: " &
VAR MaxVoteShare = MAX('ConstituencyWiseDetails'[% of Votes])
VAR SecondMaxVoteShare =
    MAXX(
        FILTER(
            'ConstituencyWiseDetails',
            'ConstituencyWiseDetails'[% of Votes] < MaxVoteShare
        ),
        'ConstituencyWiseDetails'[% of Votes]
    )
RETURN
    IF(
        ISBLANK(SecondMaxVoteShare),
        "No second max",
        SecondMaxVoteShare & " %"
    )

```

```

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=====
=====

```

```

2nd Runner Up Winning Candidate =
VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])
VAR SecondMaxVotes =
    MAXX(

```

```

    FILTER(
        'ConstituencyWiseDetails',
        'ConstituencyWiseDetails'[Total Votes] < MaxVotes
    ),
    'ConstituencyWiseDetails'[Total Votes]
)
VAR ThirdMaxVotes =
    MAXX(
        FILTER(
            'ConstituencyWiseDetails',
            'ConstituencyWiseDetails'[Total Votes] < SecondMaxVotes
        ),
        'ConstituencyWiseDetails'[Total Votes]
    )
RETURN
    CALCULATE(
        MAX('ConstituencyWiseDetails'[Candidate]),
        'ConstituencyWiseDetails'[Total Votes] = ThirdMaxVotes
    )

```

```

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=====
=====

```

2nd Runner Up Winning Party =

```
VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])
```

```

VAR SecondMaxVotes =
    MAXX(
        FILTER(
            'ConstituencyWiseDetails',
            'ConstituencyWiseDetails'[Total Votes] < MaxVotes
        ),
        'ConstituencyWiseDetails'[Total Votes]
    )
VAR ThirdMaxVotes =
    MAXX(
        FILTER(
            'ConstituencyWiseDetails',
            'ConstituencyWiseDetails'[Total Votes] < SecondMaxVotes
        ),
        'ConstituencyWiseDetails'[Total Votes]
    )
RETURN
    CALCULATE(
        MAX('ConstituencyWiseDetails'[Party]),
        'ConstituencyWiseDetails'[Total Votes] = ThirdMaxVotes
    )

```

```

=====
=====
=====

```


2nd Runner Up Winning Candidate Total Votes =

"Total Votes: " &

VAR MaxVotes = MAX('ConstituencyWiseDetails'[Total Votes])

VAR SecondMaxVotes =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[Total Votes] < MaxVotes

),

'ConstituencyWiseDetails'[Total Votes]

)

VAR ThirdMaxVotes =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[Total Votes] < SecondMaxVotes

),

'ConstituencyWiseDetails'[Total Votes]

)

RETURN

IF(

ISBLANK(ThirdMaxVotes),

"No third max",

ThirdMaxVotes

)

=====

=====

=====

2nd Runner Up Winning Candidate Vote Share % =

"Vote Share: " &

VAR MaxVoteShare = MAX('ConstituencyWiseDetails'[% of Votes])

VAR SecondMaxVoteShare =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[% of Votes] < MaxVoteShare

),

'ConstituencyWiseDetails'[% of Votes]

)

VAR ThirdMaxVoteShare =

MAXX(

FILTER(

'ConstituencyWiseDetails',

'ConstituencyWiseDetails'[% of Votes] < SecondMaxVoteShare

),

'ConstituencyWiseDetails'[% of Votes]

)

RETURN

IF(

ISBLANK(ThirdMaxVoteShare),

"No third max",

ThirdMaxVoteShare & " %"

)

9. Party Alliance Logic

Political parties in India often form alliances. To analyse alliance-wise performance, a calculated column called Party Alliance was created. This column categorizes each party into NDA, I.N.D.I.A., or OTHER.

The logic uses conditional statements to map party names to their respective alliances. This classification enables alliance-based filtering, grouping, and visualization.

The Party Alliance column is used in slicers, charts, and maps to show alliance dominance across states and constituencies.

This logic simplifies analysis by reducing complexity and improving interpretability of results.

10. Winning Alliance Calculation

The Winning Alliance measure identifies which alliance has won the highest number of seats. This is calculated using DAX measures that count the number of constituencies won by each alliance.

Using variables in DAX improves readability and performance. The CALCULATE and COUNTROWS functions are used along with alliance filters.

The measure dynamically updates based on slicers such as state selection, allowing state-wise winning alliance analysis.

This calculation is essential for high-level political insights and dashboard summaries.

11. State-Level Analysis

State-level analysis helps understand regional voting patterns and political dominance. In this project, states are used as a key dimension for analysis.

Maps and charts display seat distribution, winning alliances, and party performance at the state level. Selecting a state dynamically updates all related visuals.

This level of analysis highlights regional differences and shows how political preferences vary across India.

12. State Demographics Dashboard

The state demographics dashboard provides a visual summary of election results for each state. It includes maps, cards, and charts showing total seats, winning alliance, and party-wise distribution.

Interactive elements allow users to explore individual states in detail. This dashboard enhances user engagement and improves data understanding.

The design focuses on clarity, consistency, and ease of navigation.

13. Constituency Analysis

Constituency-level analysis provides granular insights into election outcomes. Each constituency represents a unique electoral contest with multiple candidates.

This analysis identifies winners, runner-ups, and vote margins. Tables and tooltips are used to display detailed candidate information.

Constituency analysis is crucial for understanding close contests and local voting behaviour.

14. Winning Candidate Logic

The winning candidate is identified based on the highest number of votes in a

constituency. DAX measures are used to calculate the maximum votes and retrieve the corresponding candidate and party.

This logic ensures accuracy even when filters are applied. It forms the basis for winner cards and detailed views.

Winning candidate identification is a core component of the dashboard.

15. Runner-Up & 2nd Runner-Up Logic

Runner-up and second runner-up candidates are identified using DAX functions like `MAXX` and `FILTER`. These calculations help analyze electoral competitiveness.

Understanding runner-up performance provides insights into close elections and potential swing constituencies.

This logic enhances analytical depth beyond just winners.

16. Vote Share & Margin Analysis

Vote share represents the percentage of total votes received by a candidate.

Margin analysis calculates the difference between winner and runner-up votes.

These metrics highlight the strength of victory and competitiveness of constituencies. High margins indicate strong support, while low margins indicate close contests.

Vote share and margin analysis add significant value to election insights.

17. Interactivity & Slicers

Interactivity is a key strength of Power BI. Slicers allow users to filter data by state, party, alliance, and constituency.

Cross-filtering ensures that selections in one visual update all others. This makes exploration intuitive and dynamic.

Interactive dashboards improve user experience and analytical flexibility.

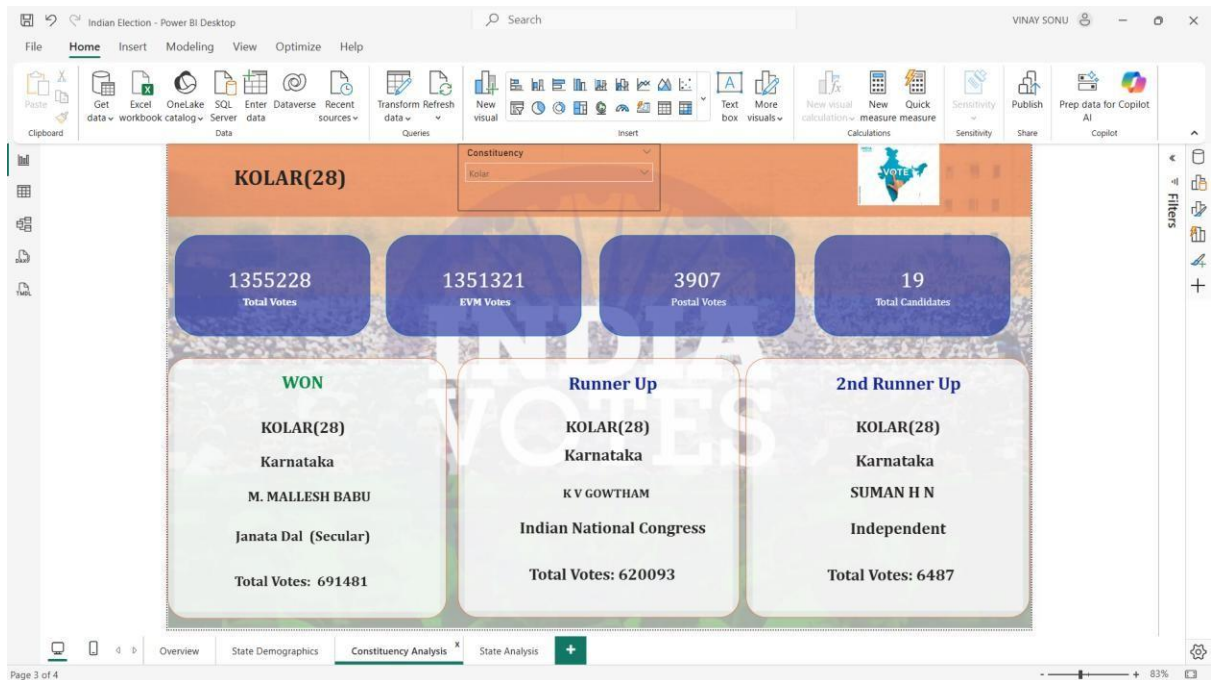
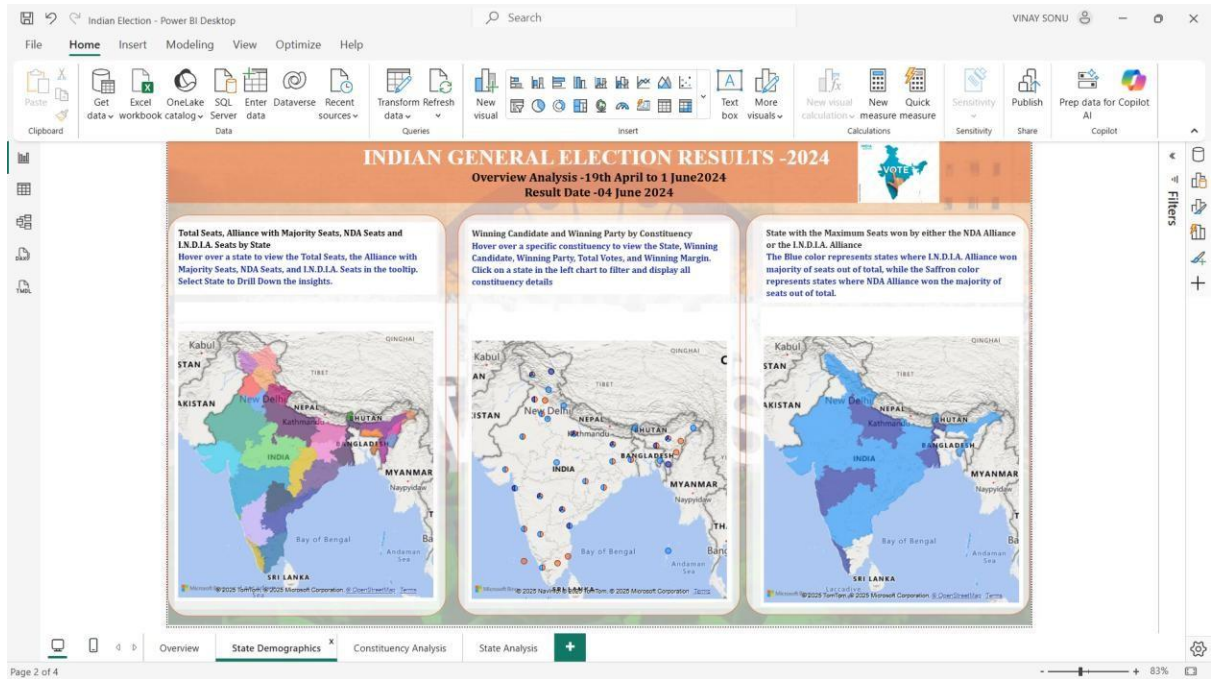
18. Visualization

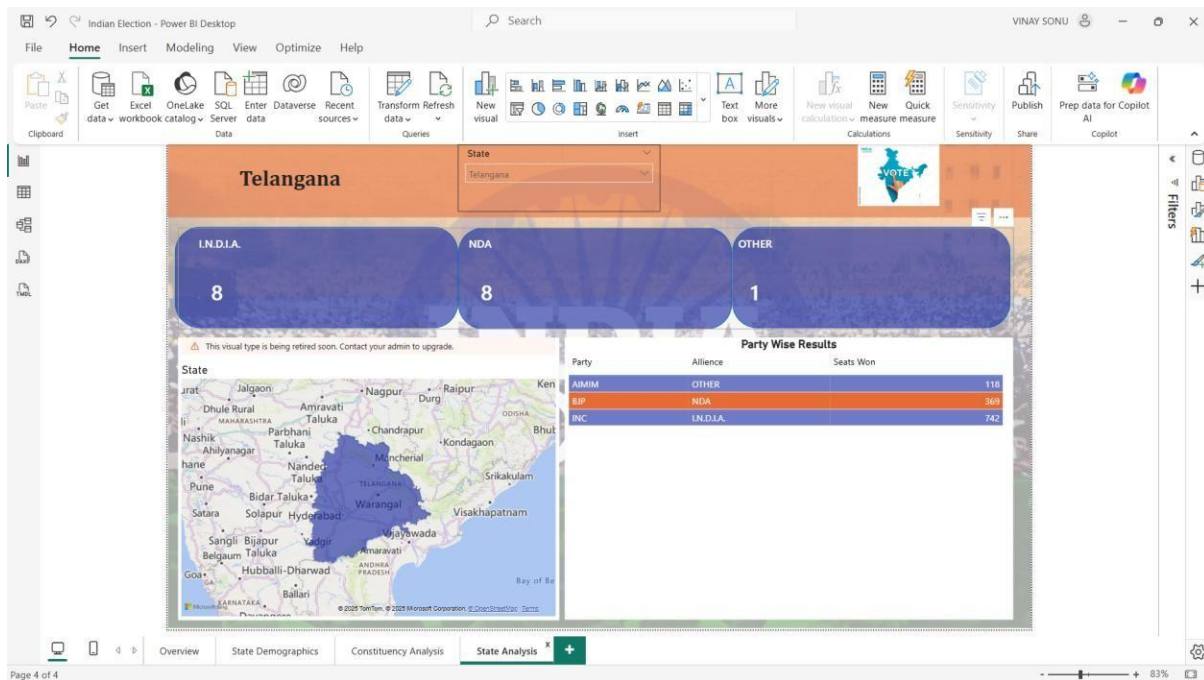
Effective visualization is essential for clear communication. This project uses maps, cards, tables, and charts with consistent color coding.

Alliance colors are used for easy identification. Layout and spacing are designed to reduce clutter and improve readability.

Good UX design ensures that insights are accessible to both technical and non-technical users.







19. Key Insights

The dashboard reveals several insights, such as alliance dominance, state-wise variations, and close electoral contests.

Certain states show clear alliance preferences, while others display competitive multi-party contests. Vote margins vary significantly across constituencies.

These insights demonstrate the power of data visualization in understanding complex election data.

20. Conclusion & Future Enhancements

This project successfully demonstrates how Power BI can be used to analyse large-scale election data. By combining data modelling, DAX, and visualization, meaningful insights were generated.

Future enhancements may include gender analysis, age-wise candidate analysis, historical election comparison, and predictive analytics.

Overall, this project highlights the importance of data analytics in public decision-making and governance.

